

Phase transitions in solutions

Y16 (2016) — English translation

Liquids can transform into solid and gaseous states - first-order phase transitions (that is, transitions associated with the absorption or release of heat). These transitions occur at constant temperatures. In tables, these temperatures are usually provided for normal conditions. But their meaning depends on many factors. In this problem it is proposed to consider some of them (only first-order phase transitions are considered).

The phase transition curve of a substance from the 1 state to the 2 state is described by the Clapeyron - Clausius equation

$$\frac{dp}{dT} = \frac{q_{12}}{T(v_2 - v_1)}$$

где p — давление, T — температура, q_{12} — удельная теплота фазового перехода $1 \rightarrow 2$, v_1, v_2 — specific volumes.

- | | | |
|-----------|--|-------------|
| A1 | At what temperature does water boil in Switzerland at Dufour Peak at an altitude of 4634 м, if the pressure there is 430 мм.рт.ст. Normal pressure is 760 мм.рт.ст. Under normal conditions, water boils at 100°C. | 0.20 |
|-----------|--|-------------|

It turns out that the phenomenon of surface tension σ of a liquid can affect the saturated vapor pressure of a substance above a curved surface (K is the curvature of the surface, the reciprocal of the radius of curvature).

- | | | |
|-----------|--|-------------|
| A2 | Find the Laplace pressure in the case of a cylindrical surface - the difference in pressure on opposite sides of the curved surface of the liquid. Express your answer in terms of σ , K , p_a (atmospheric pressure) and physical constants. | 0.10 |
|-----------|--|-------------|

- | | | |
|-----------|--|-------------|
| A3 | Find the Laplace pressure for a spherical surface. Express your answer in terms of σ , K , p_a (atmospheric pressure) and physical constants. | 0.10 |
|-----------|--|-------------|

- | | | |
|-----------|---|-------------|
| A4 | Consider a bubble in a liquid. Where is the pressure greater: inside (IN) of the bubble or outside (OUT)? | 0.10 |
|-----------|---|-------------|

In the next part of the problem, consider how the curvature of the liquid surface affects the vapor pressure. To solve, consider communicating cylindrical vessels placed in a thermostat maintaining a temperature T , (Fig. 1 - shows the case when the liquid wets the walls of the vessels) such that the width of one of the vessels allows the capillary effect to be observed. Vapor pressure above a flat surface AB - p_1 , above CD - p_2 , K - surface curvature CD . Specific volume of liquid v_L , vapor v_G (consider that the vapor density changes slightly as the height changes by h). Rice. 1. Saturated vapor pressure over a curved surface

In the next part of the problem, consider how the curvature of the liquid surface affects the vapor pressure. To solve, consider communicating cylindrical vessels placed in a thermostat maintaining a temperature T , (Fig. 1 - shows the case when the liquid wets the walls of the vessels) such that the width of one of the vessels allows the capillary effect to be observed. Vapor pressure above a flat surface AB - p_1 , above CD - p_2 , K - surface curvature CD . Specific volume of liquid v_L , vapor v_G (consider that the vapor density changes slightly as the height changes by h).

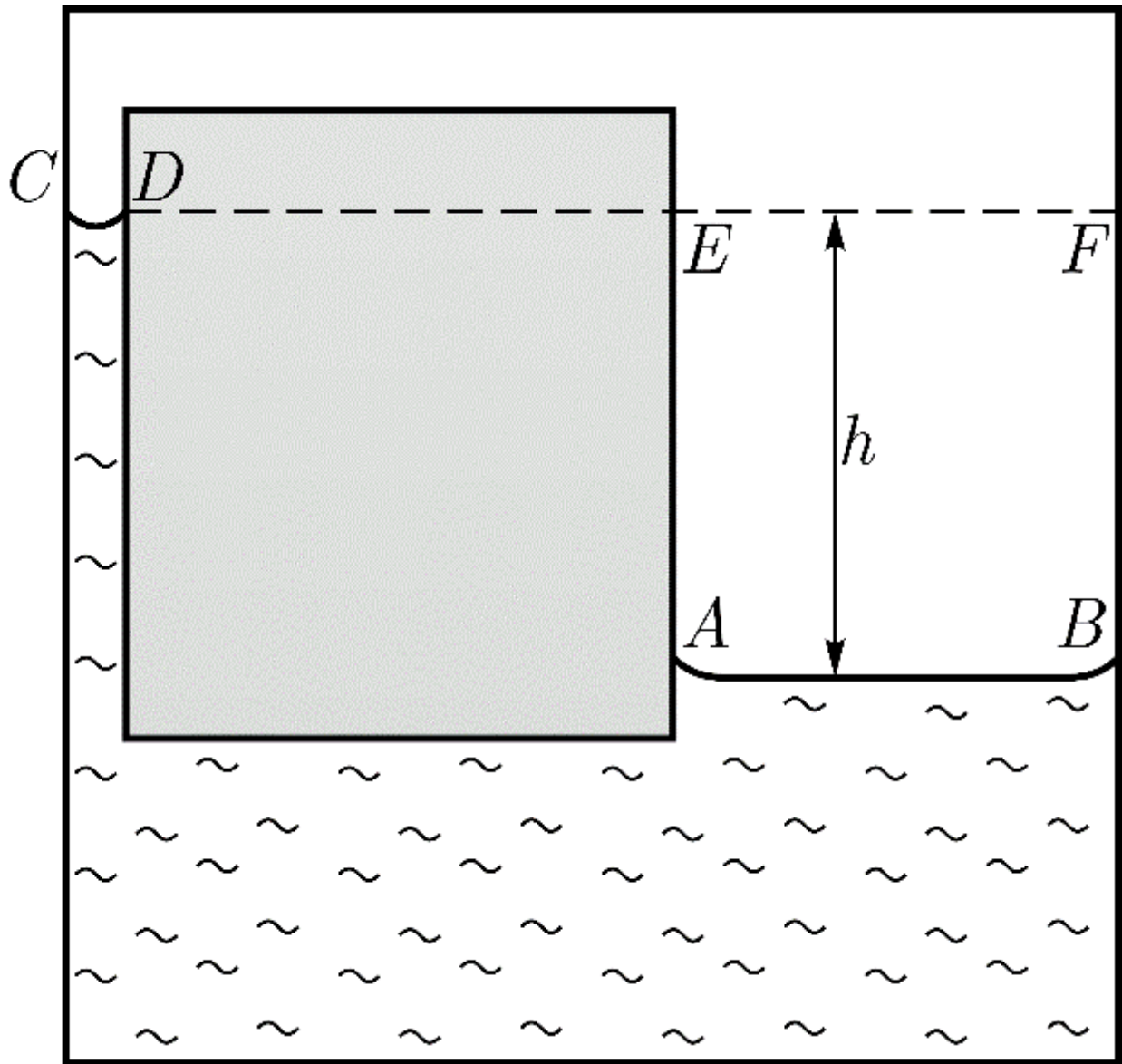


Fig. 1.

A5	Express h in terms of σ , K , v_L , v_G , T , μ and physical constants (μ is the molar mass).	0.20
A6	Let the liquid wet the walls of the vessel. Express the pressure difference $p_2 - p_1$ in terms of σ , K , v_L , v_G , T , μ and physical constants.	0.40
A7	What will be the answer if the liquid does not wet the walls of the vessel? Express your answer in terms of σ , K , v_L , v_G , T , μ and physical constants.	0.10
A8	With large curvature ($\sigma K \gg p_2 - p_1 $) of the surface, the vapor density can no longer be considered constant. What will be the pressure difference $p_2 - p_1$ in this case? Express your answer in terms of σ , K , v_L , v_G , T , μ and physical constants.	0.40

In the next part of the problem, it is proposed to consider how the conditions of phase transitions will change if any substance is dissolved in a liquid. The first thing to consider is osmosis - the process of one-way diffusion through a semi-permeable membrane. This membrane allows only solvent molecules to pass through. Solvent molecules diffuse to where the solute concentration is greater. Concentration of solvent molecules n , solute n_s . As a result of this phenomenon, osmotic pressure p_{os} arises - excess pressure in the

solution, at which the diffusion of solvent molecules stops. In Fig. Figure 2 shows an osmometer, a device for measuring osmotic pressure. Rice. 2. Osmometer

In the next part of the problem, it is proposed to consider how the conditions of phase transitions will change if any substance is dissolved in a liquid. The first thing to consider is osmosis - the process of one-way diffusion through a semi-permeable membrane. This membrane allows only solvent molecules to pass through. Solvent molecules diffuse to where the solute concentration is greater. Concentration of solvent molecules n , solute n_s . As a result of this phenomenon, osmotic pressure p_{os} arises - excess pressure in the solution, at which the diffusion of solvent molecules stops. In Fig. Figure 2 shows an osmometer, a device for measuring osmotic pressure.

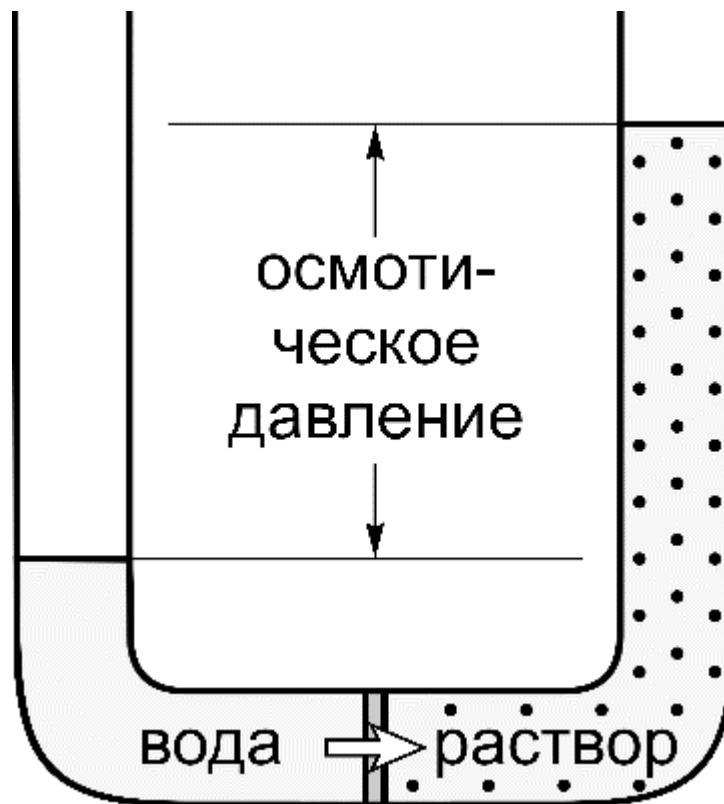


Fig. 2.

B1 Express the osmotic pressure p_{os} for a weak solution (a solution with a low concentration of solute). Express your answer in terms of p_a , n , n_s , T and physical constants. **0.50**

Consider how the saturated vapor pressure changes if a small amount of a substance is dissolved in a liquid. In Fig. 3, the solution and water are separated by a semi-permeable membrane, p_0 is the saturated vapor pressure above the water, p is above the solution.

Consider how the saturated vapor pressure changes if a small amount of a substance is dissolved in a liquid. In Fig. 3, the solution and water are separated by a semi-permeable membrane, p_0 is the saturated vapor pressure above the water, p is above the solution.

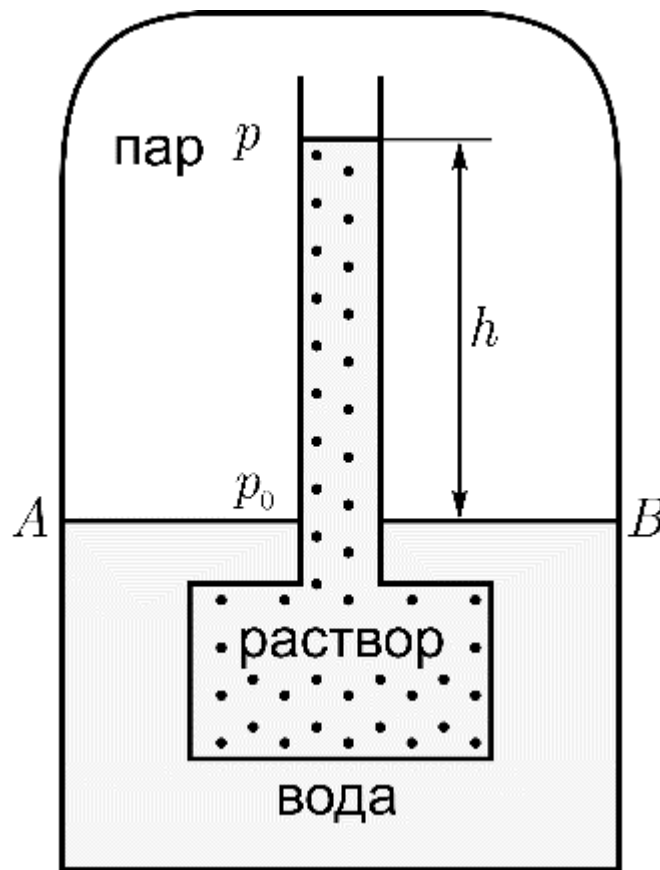


Fig. 3.

B2 Express the difference $p - p_0$ in terms of v_L , v_G , p_{os} and physical constants. **0.50**

B3 Express the difference $p - p_0$ in terms of p_0 , n , n_s and physical constants. **0.30**

In the next part of the problem, consider three phases of matter (1 - liquid, 2 - vapor, 3 - solid, q_{21} , q_{23} , q_{13} - specific heats released during transitions $2 \rightarrow 1$, $2 \rightarrow 3$ and $1 \rightarrow 3$, respectively, $q_{21} > 0$, $q_{23} > 0$, $q_{13} > 0$). Consider the three-phase equilibrium point (triple point).

C1 Assuming the difference between p and p_0 to be small, find the change in the boiling point of the liquid when a substance is dissolved in it. Express your answer in terms of μ , T , q_{12} , n_s , n and physical constants. **2.00**

C2 What is the sign of the change in boiling point of a liquid when something is dissolved in it? **0.30**

C3 Express q_{23} in terms of q_{13} and q_{21} at the triple point. **1.50**

C4 In coordinates (p, T) , draw qualitatively the equilibrium curves between all three phases near the triple point. Consider $v_G \gg v_L$ and $v_G \gg v_s$ (specific volume of solid). **1.00**

C5 Find the change in the freezing point of a liquid when a substance is dissolved in it. Express your answer in terms of μ , T , q_{13} , n_s , n and physical constants. **2.00**

C6 What is the sign of the change in the freezing point of a liquid when something is dissolved in it? **0.30**

Source: <https://pho.rs/p/3028>